Homework 4 – Regression Consult

When people make decisions, their choices are often swayed by how information is presented. This is often described as "the framing effect." One example of this, is how commercials describe new medications. As an example, Nereus (tradipitant), a new drug for motion-induced vomiting like sea-sickness, is marketed with a **positive frame** as reducing vomiting incidence from 44% to 18% (a reduction of 60%). This marketing makes it seem more effective than a **negative frame** that says just 18% of patients still experienced vomiting, even though both statements are true.

Xu (2024) aimed to evaluate whether older adults are more or less vulnerable to framing than younger adults. They collected a sample of 696 adults aged 18 to 71 and exposed them to three different scenarios

Scenario 1: Rate Drug Efficacy (1 = very bad, 6 = very good)

- **Positive Frame:** 70 out of 100 patients felt better
- **Negative Frame:** 30 out of 100 patients did not feel better

Scenario 2: Likelihood to stop drinking cocoa (1 = surely stop, 6 = surely continue)

- **Positive Frame:** Stopping drinking hot cocoa reduces cadmium and cancer risk.
- **Negative Frame:** Continuing to drink hot cocoa increases cadmium and cancer risk.

Scenario 3: Preference for chance choice (1 = surely safe option, 6 = surely risky)

- **Positive Frame:** Choice between 200 saved vs 1/3 chance all 600 are saved.
- **Negative Frame:** Choice between 400 die vs 2/3 chance all 600 die.

In each scenario, participants were randomly assigned a positive or negative frame. They rated their response on a 1-6 scale. The researchers want to build three models, one for each scenario adjusting for simultaneous linear changes in demographic variables (gender, ethnicity, education level, political affiliation).

They want to evaluate whether, as adults age, the way choices are worded have more influence on their judgments and ultimate choices. If there is a change, are those changes driven by positive or negative framings? At what age does this effect onset?

Your Goals:

- Determine the correct analysis strategy and conduct the necessary analyses.
  - Hint 1: Ensure to assess and communicate results about the model fit.
  - Hint 2: Ensure to think about whether or which interactions need to be included.
  - Hint 3: You will need run various post hoc procedures (e.g., `emmeans`, `emtrends` or `johnson_neyman`).
  - Hint 4: Looking at the paper may be helpful. They do include `R` code, but I expect you to do a much better job using the techniques we discussed (they do somethings differently)
- Write a clear, concise, complete, and correct report conveying an answer to their research question.

The data are in `HW4Data.csv` but they will require some manual and `tidyverse` cleaning.

1 Loading and Cleaning the Data

Solution

Xu (2024) stated in the paper that they already removed 153 participants who did not pass the attention check and 1 response with mismatched age, leaving us with 696 respondents (out of 850 that initially responded). I transformed the categorical variable `education` to ordinal, and since "Less than high school" only has 2 responses, I collapsed it

with "High school graduate" to become "1". Similarly, I collapse the levels with not a lot of responses in `gender`, `ethnicity` and `politics`. Age is centered.

```
dat = read.csv("HW4Data.csv")
```

```
dat = dat |>
  select(age, gender, politics, ethnicity, education,
         treatment1, treatment2, treatment3,
         pick1, pick2, pick3) |>
  mutate(education = case_when(education == "Less than high school" ~ "1",
                               education == "High school graduate" ~ "1",
                               education == "Some college" ~ "2",
                               education == "2 year degree" ~ "3",
                               education == "4 year degree" ~ "4",
                               education == "Professional degree" ~ "5",
                               education == "Doctorate" ~ "6",
                               TRUE ~ education)) |>
  mutate(gender = case_when(gender == "Non-binary / third gender" ~ "Third gender/Prefer not to say",
                            gender == "Prefer not to say" ~ "Third gender/Prefer not to say",
                            TRUE ~ gender)) |>
  mutate(ethnicity = case_when(ethnicity == "American Indian or Alaska Native"  ~ "Other/Prefer not to say",
                               ethnicity == "Native Hawaiian or Pacific Islander"  ~ "Other/Prefer not to say",
                               ethnicity == "Other" ~ "Other/Prefer not to say",
                               ethnicity == "Prefer not to say" ~ "Other/Prefer not to say",
                               TRUE ~ ethnicity)) |>
  mutate(politics = case_when(politics == "Prefer not to say"  ~ "Other/Prefer not to say",
                              politics == "Something else"  ~ "Other/Prefer not to say",
                              TRUE ~ politics)) |>
  mutate(age_c = age - mean(age, na.rm = TRUE))
```

2 Exploratory Data Analyses

Solution

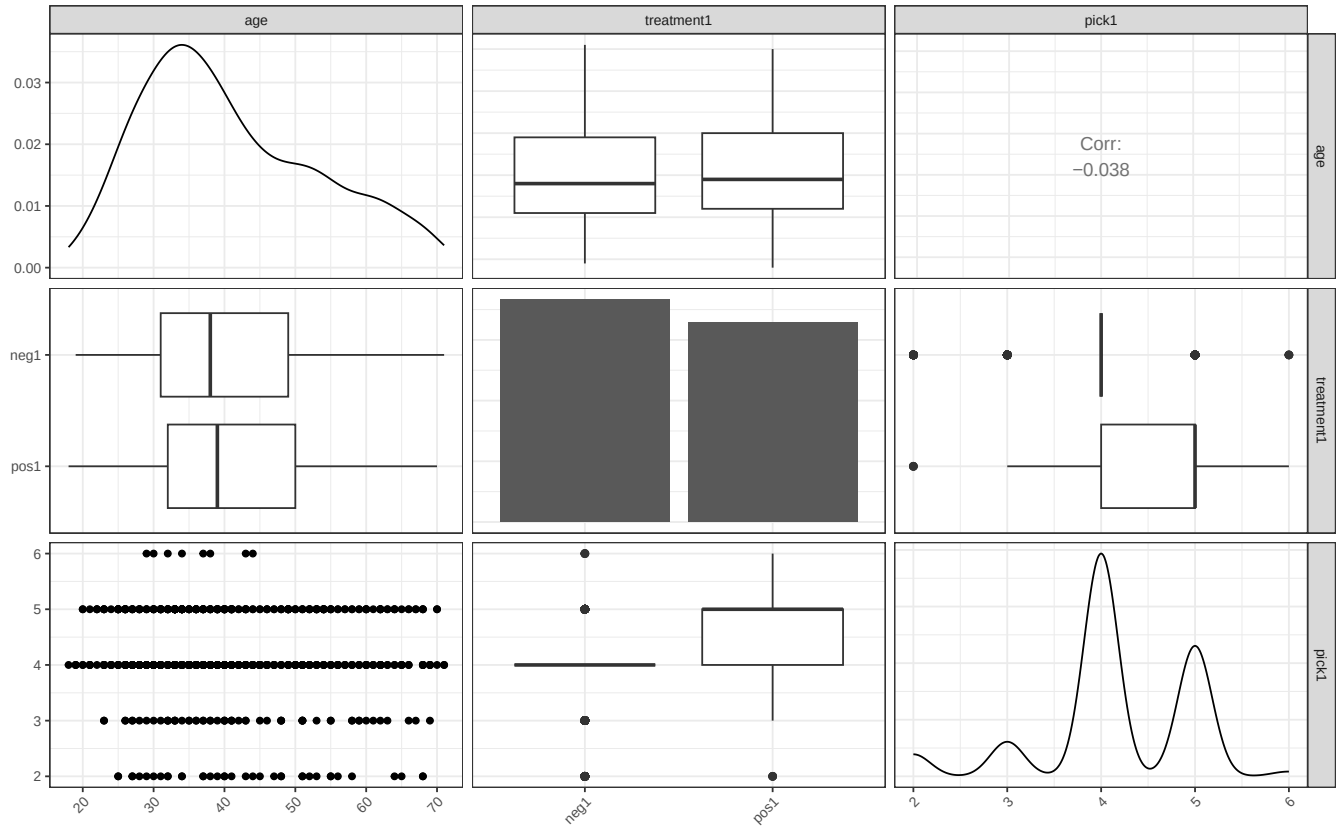
Figure 1, Figure 2 and Figure 3 demonstrate that age is approximately normally distributed. The distribution is right-skewed, peaking around 30, showing that our sample comprises of mostly young and middle-age people. The treatment groups are roughly equal in size in all 3 experiments.

The correlation between average scenario rating (i.e., pick) and age is not significant, but this doesn't rule out the possibility of the effect of the Treatment $\times$ Age interaction.

Looking at Figure 4, we see that an overwhelming majority of our sample are White. The sample consists of more female than male. A majority have high education level (4-year-degree or more), and identify as democrats or independent. The sample is therefore not quite representative of the general population that the study targets.

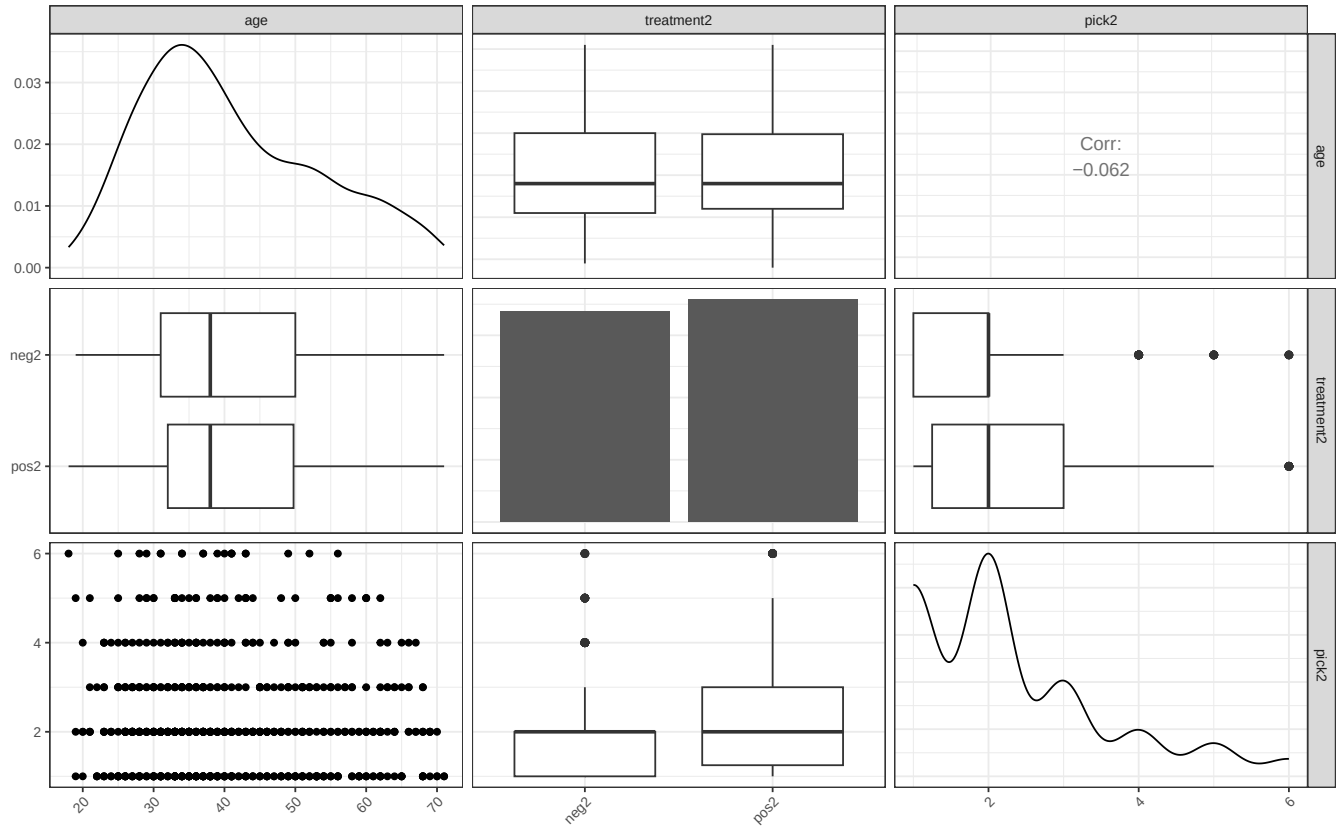
```
ggpairs(dat,
        columns = c("age", "treatment1", "pick1"),
        lower = list(combo = "box_no_facet", discrete = "count"),
        upper = list(combo = "box_no_facet", discrete = "count"),
        diag  = list(discrete = "barDiag")) +
  theme_bw(base_size = 8) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

Figure 1: Pairwise Plots for Attribute Framing



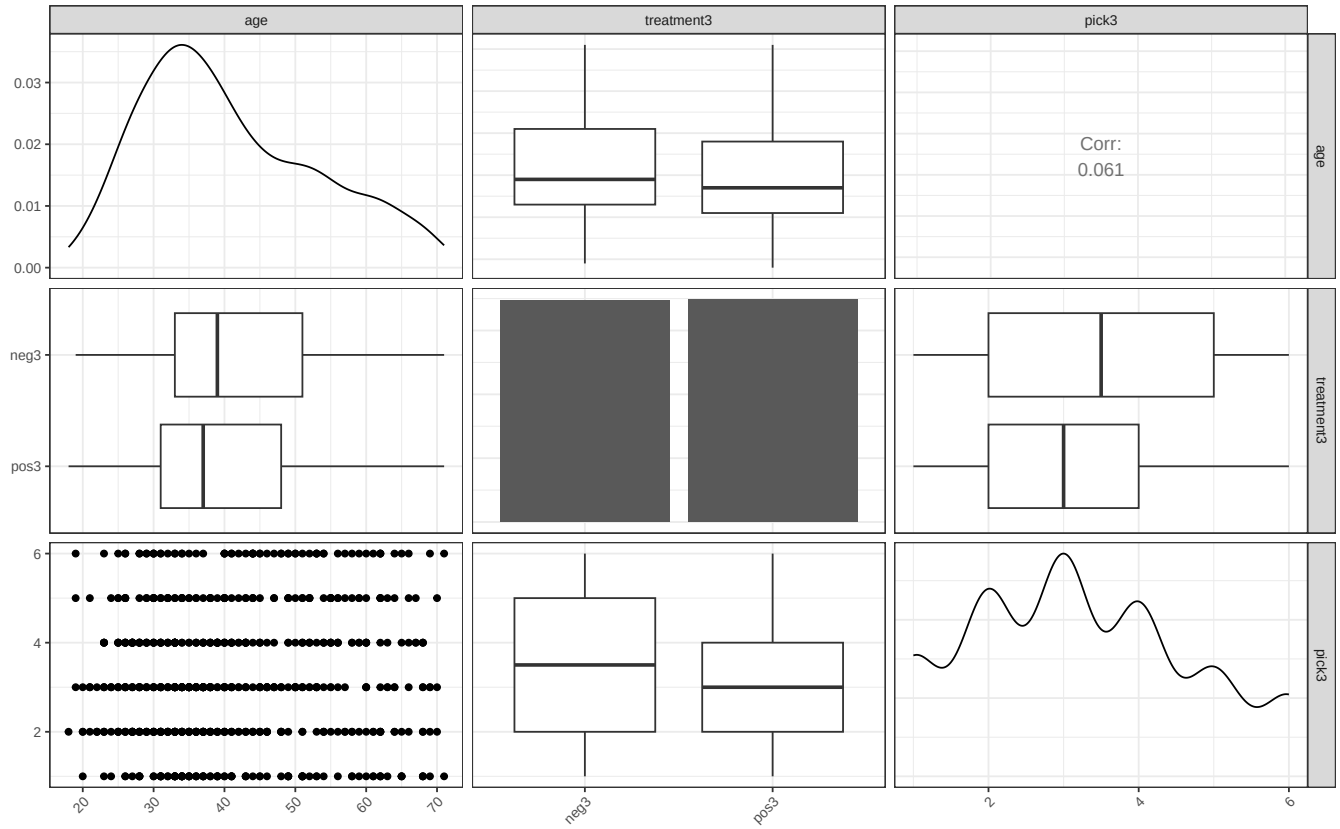

```
ggpairs(dat,
        columns = c("age", "treatment2", "pick2"),
        lower = list(combo = "box_no_facet", discrete = "count"),
        upper = list(combo = "box_no_facet", discrete = "count"),
        diag  = list(discrete = "barDiag")) +
  theme_bw(base_size = 8) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

Figure 2: Pairwise Plots for Goal Framing



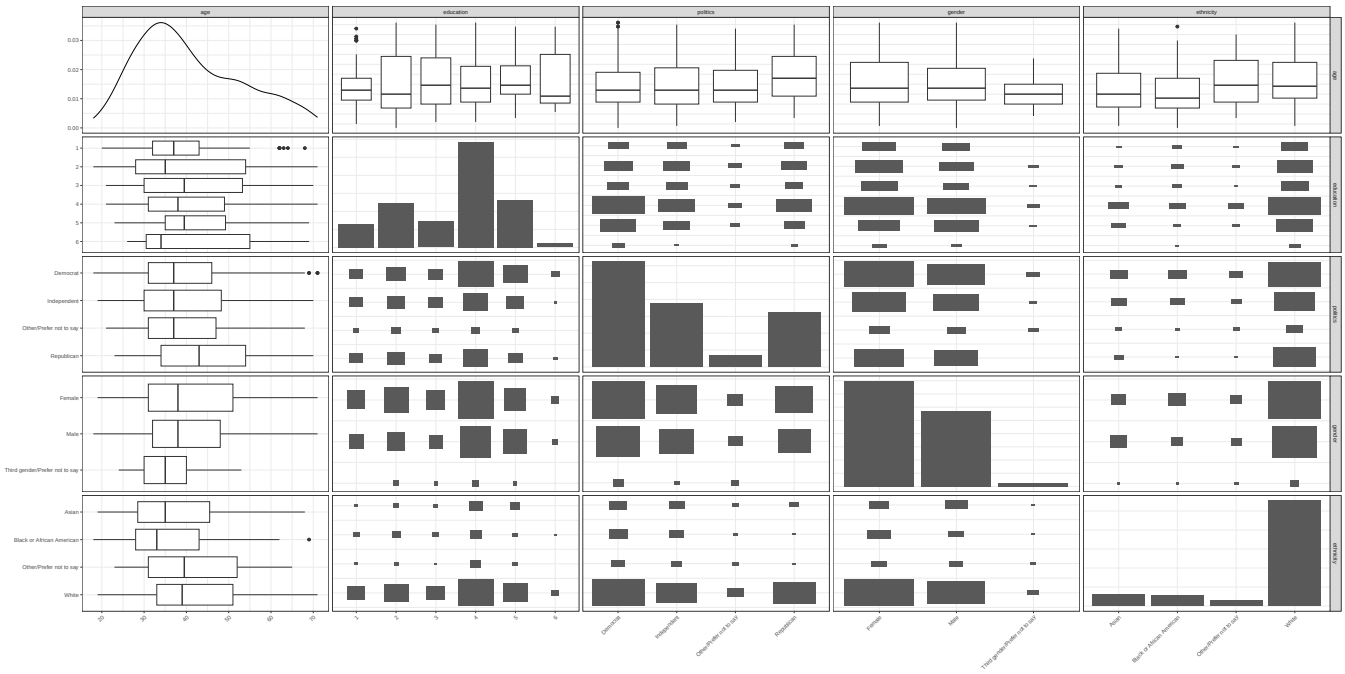

```
ggpairs(dat,
        columns = c("age", "treatment3", "pick3"),
        lower = list(combo = "box_no_facet", discrete = "count"),
        upper = list(combo = "box_no_facet", discrete = "count"),
        diag  = list(discrete = "barDiag")) +
  theme_bw(base_size = 8) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

Figure 3: Pairwise Plots for Risky Choice Framing




```
ggpairs(dat,
        columns = c("age", "education", "politics", "gender", "ethnicity"),
        lower = list(combo = "box_no_facet", discrete = "count"),
        upper = list(combo = "box_no_facet", discrete = "count"),
        diag  = list(discrete = "barDiag")) +
  theme_bw(base_size = 8) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

Figure 4: Pairwise Plots for Age and Demographics



3 Regression Analyses

```
source("plotResiduals.R")
source("plotInfluence.R")
```

3.1 Rating Drug Efficacy

Solution

First, we fit the regression model, then conduct residual and outlier analysis and multicolinearity check to make sure the conditions for use are met.

According to Figure 5, our residuals follow an approximate normal distribution, are centered around 0, and have common variance. The reason why we see multiple peaks in the qq plot and stripes in the fitted values vs residual plot is because the response is from a 6-point Likert scale. For example, one of the stripe can be written as:

$$Residual = 6 - \hat{Y}$$

According to Figure 6, we have many observations with high leverage, but Cook's D graph shows that they're not all influential. A few outliers can be identified from the Studentized Residuals and DFFITs graphs. Table 1 indicates that there's no multicolinearity. Overall, our data meets the conditions for use of regression.

The model discernibly explains the variation in the outcome, $F_{17,679} = 10.41, p < 0.001, \omega^2 = 0.18$, see Table 2. This effect size is medium.

Table 3 shows that the effect of treatment is statistically discernible, $F_{1,679} = 141.65, p < 0.001$, but the effect of the interaction between treatment and age is not, $F_{1,679} = 3.17, p = 0.08$.

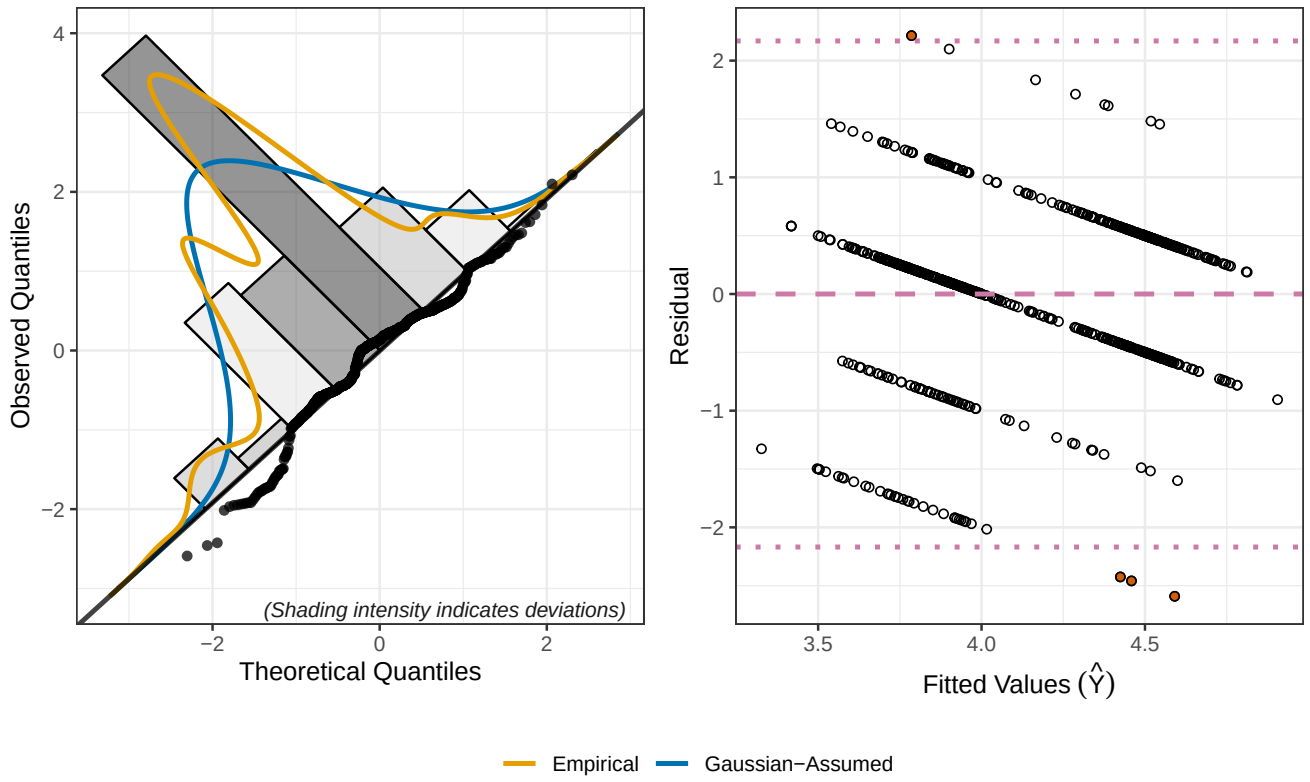
3.1.1 Fit the model

```
mod.q1 = lm(pick1 ~ treatment1 * age_c + gender + ethnicity
                    + education + politics,
                    data = dat)
```

3.1.2 Residual and outlier analysis

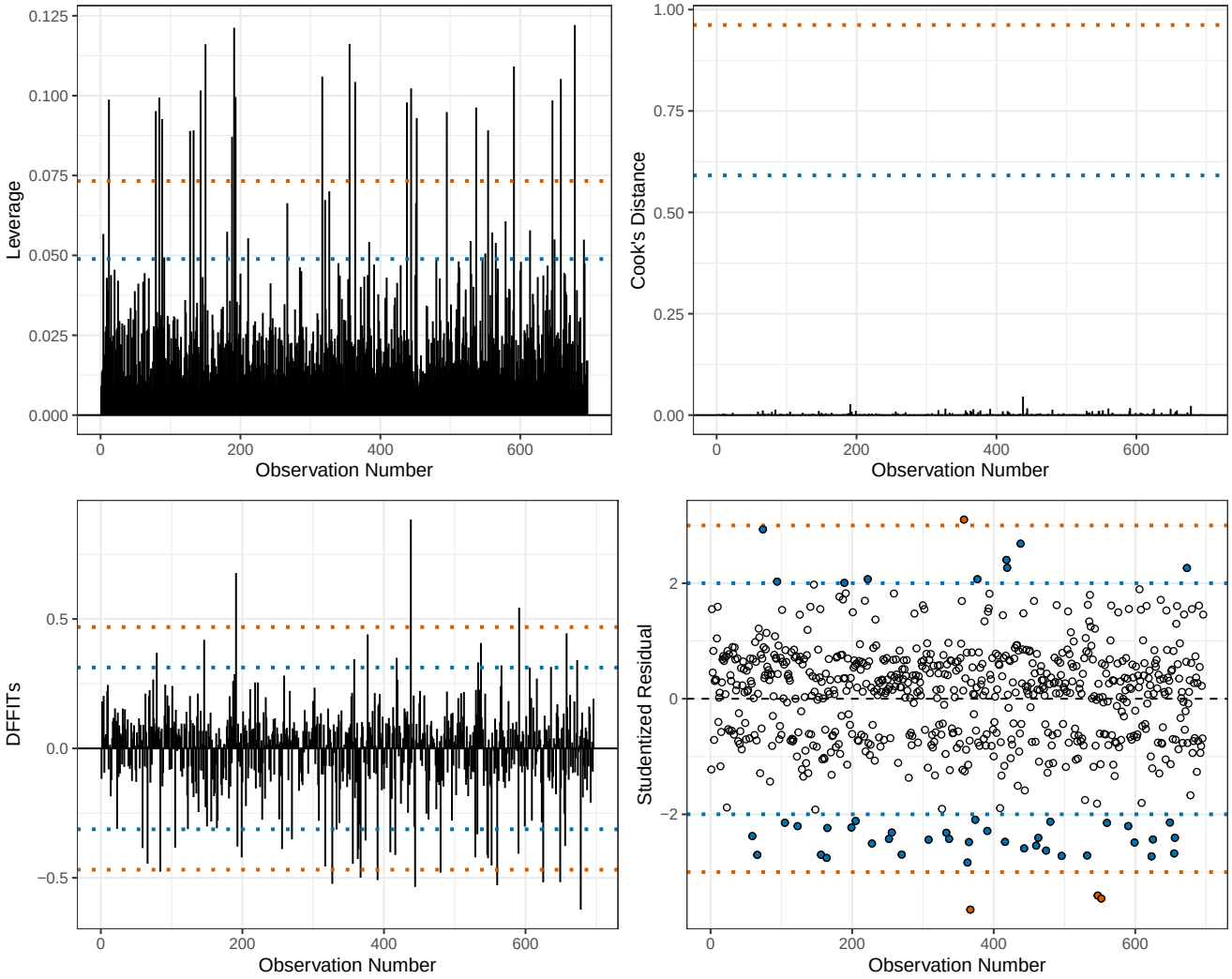
```
plotResiduals(mod.q1)
```

Figure 5: Residual Analysis




```
plotInfluence(mod.q1)
```

Figure 6: Outlider Analysis



3.1.3 Multicolinearity check

Table 1: VIF

	GVIF	Df	GVIF^(1/(2*Df))
treatment1	1.0254	1	1.0126
age_c	1.8712	1	1.3679
gender	1.0810	2	1.0197
ethnicity	1.1281	3	1.0203
education	1.1406	5	1.0132
politics	1.1763	3	1.0274
treatment1:age_c	1.8637	1	1.3652

3.1.4 Overall F-test

```
sum.q1 = summary(mod.q1)
```

```
q1 = as_tibble_row(sum.q1$fstatistic) |>
  rename(F = value, df1 = numdf, df2 = dendf) |>
  mutate(p = pf(F, df1, df2, lower.tail = FALSE),
         omega2 = F_to_omega2(F, df1, df2)$Omega2)
```

```
(interpret_omega_squared(0.18, "cohen1992"))
```

```
[1] "medium"
(Rules: cohen1992)
```

Table 2: Overall F test

F	df1	df2	p	omega2
10.4059	16	679	0	0.1778

Table 3: ANOVA

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
treatment1	1	74.0112	74.0112	141.6525	0.0000
age_c	1	0.9116	0.9116	1.7448	0.1870
gender	2	0.5900	0.2950	0.5647	0.5688
ethnicity	3	4.2377	1.4126	2.7036	0.0446
education	5	3.9259	0.7852	1.5028	0.1867
politics	3	1.6551	0.5517	1.0559	0.3672
treatment1:age_c	1	1.6588	1.6588	3.1749	0.0752
Residuals	679	354.7668	0.5225	NA	NA

3.2 Likelihood to Stop Drinking Cocoa

First, we fit the model. A residual analysis, as in Figure 7, reveals that while the residuals are approximately normal, there are more positive than negative residuals. I performed a log transformation to fix this problem, shown in Figure 8. Here, although the points towards the left tail are still off the qq line and we have more residuals above 0 than below, the data is now more centered around 0.

Figure 9 reveals 2 or 3 observations that both have high leverage and are influential (DFFITs). However, Studentized Residual and Cook's D plots shows that they are only moderate outliers. Table 4 shows that there is no significant multicolinearity. The conditions for use of regression are generally met.

The model discernibly explains the variation in the outcome, $F_{16,679} = 2.86, p < 0.001, \omega^2 = 0.04$. The overall effect size is small.

Results are for $log(rating)$. Table 6 shows that the effect of treatment is statistically discernible, $F_{1,679} = 26.71, p < 0.001$, but the effect of the interaction between treatment and age is not, $F_{1,679} = 3.32, p = 0.070$.

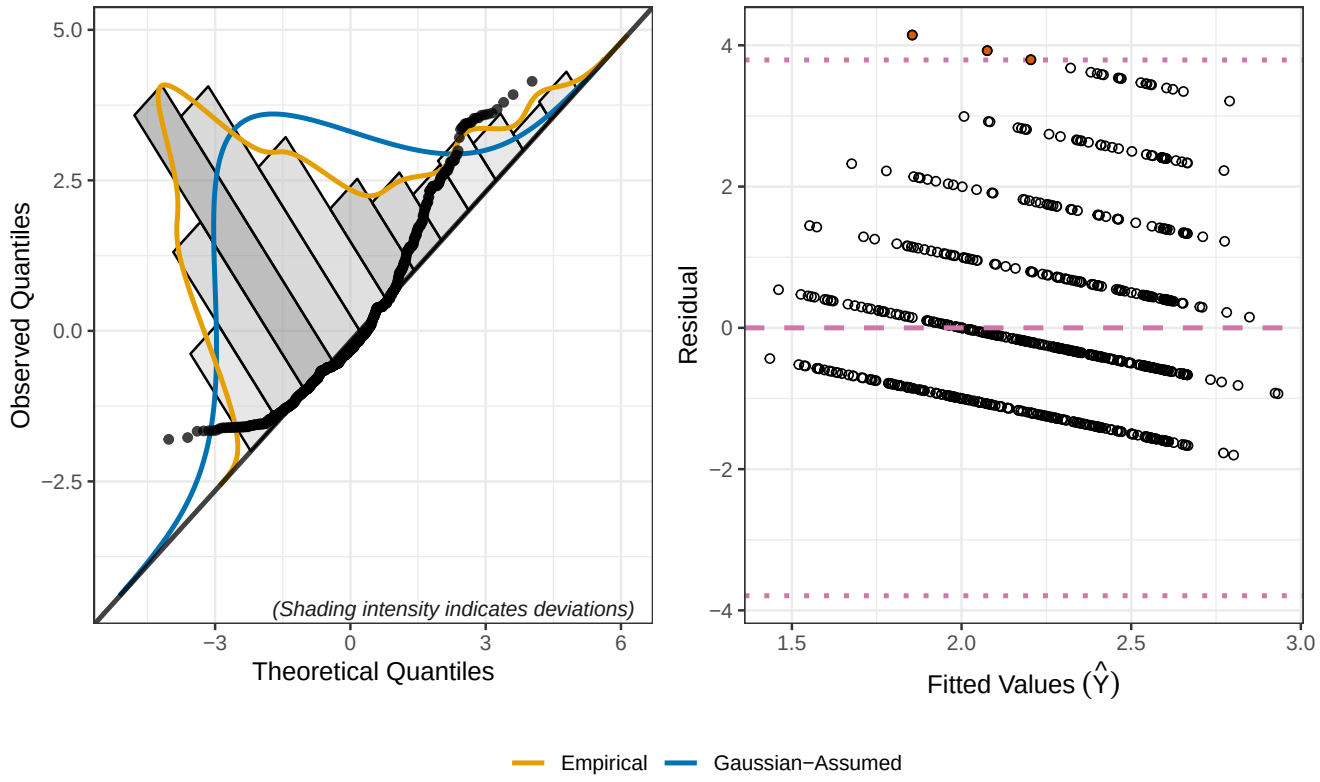
3.2.1 Fit the model

```
mod.q2 = lm(pick2 ~ treatment2 * age_c + gender + ethnicity
                    + education + politics,
                    data = dat)
```

3.2.2 Residual and outlier analysis

```
plotResiduals(mod.q2)
```

Figure 7: Residual Analysis

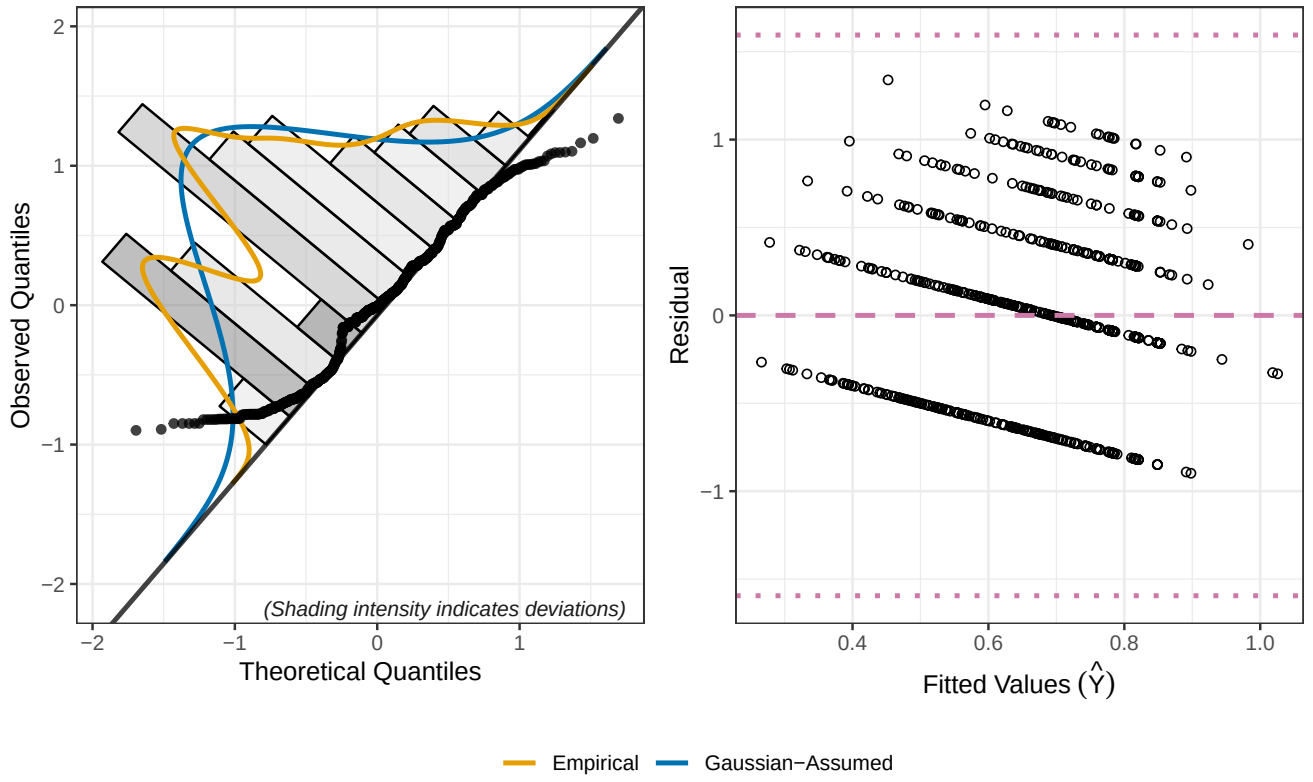


3.2.3 Log transformation

```
mod.q2.log = lm(log(pick2) ~ treatment2 * age_c + gender + ethnicity
                    + education + politics,
                    data = dat)
```

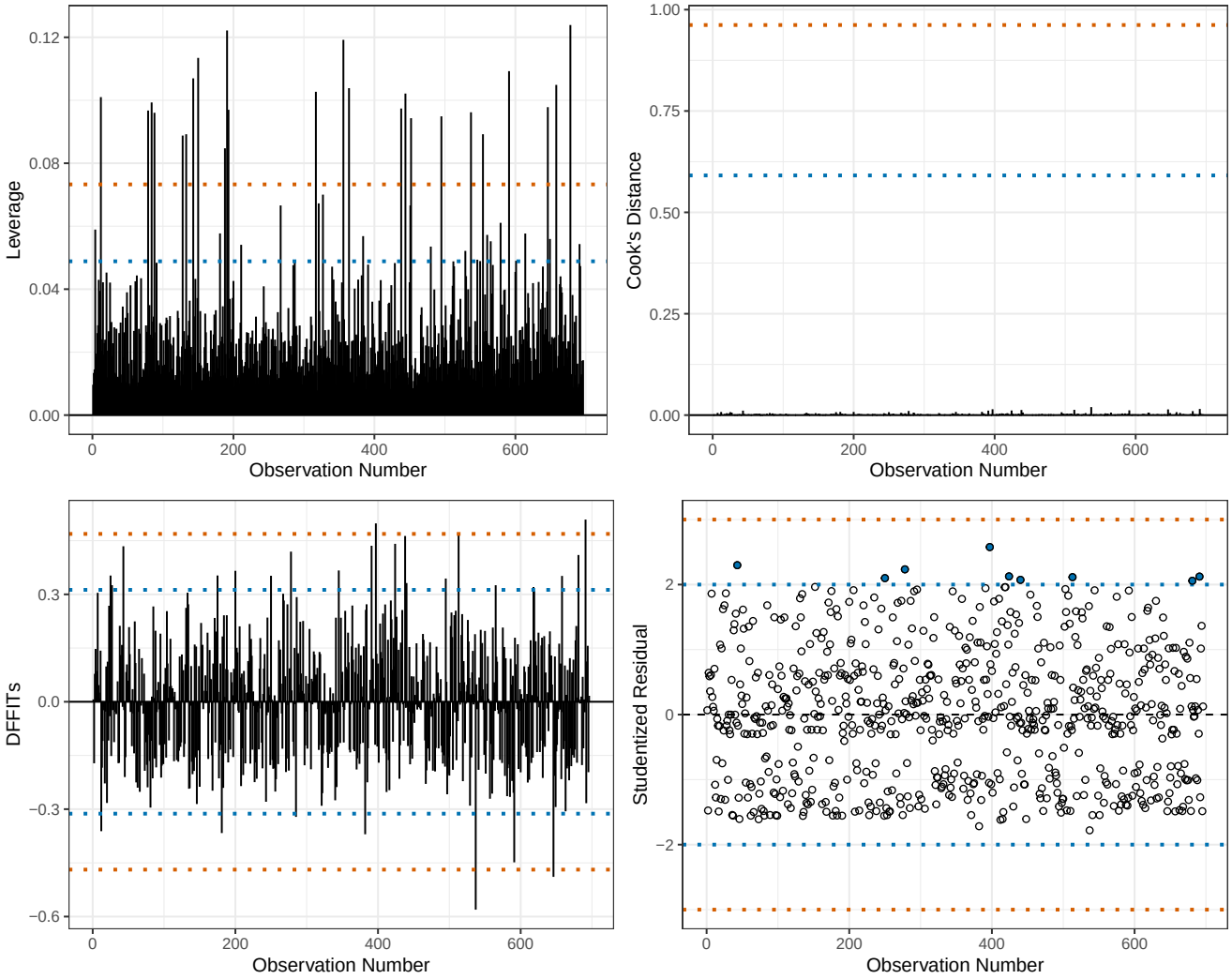
```
plotResiduals(mod.q2.log)
```

Figure 8: Residual Analysis After Log Transformation




```
plotInfluence(mod.q2.log)
```

Figure 9: Outlider Analysis



3.2.4 Multicolinearity check

Table 4: VIF

	GVIF	Df	GVIF^(1/(2*Df))
treatment2	1.0262	1	1.0130
age_c	2.0627	1	1.4362
gender	1.0794	2	1.0193
ethnicity	1.1293	3	1.0205
education	1.1497	5	1.0140
politics	1.1645	3	1.0257
treatment2:age_c	2.0407	1	1.4285

3.2.5 Overall F-test

```
sum.q2 = summary(mod.q2.log)
```

```
q2 = as_tibble_row(sum.q2$fstatistic) |>
  rename(F = value, df1 = numdf, df2 = dendf) |>
  mutate(p = pf(F, df1, df2, lower.tail = FALSE),
         omega2 = F_to_omega2(F, df1, df2)$Omega2)
```

```
(interpret_omega_squared(0.04, "cohen1992"))
```

```
[1] "small"
(Rules: cohen1992)
```

Table 5: Overall F test

F	df1	df2	p	omega2
2.8615	16	679	2e-04	0.041

Table 6: ANOVA

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
treatment2	1	7.5481	7.5481	26.7108	0.0000
age_c	1	0.9126	0.9126	3.2294	0.0728
gender	2	0.5009	0.2504	0.8863	0.4127
ethnicity	3	0.5896	0.1965	0.6954	0.5550
education	5	2.0671	0.4134	1.4630	0.1998
politics	3	0.3830	0.1277	0.4518	0.7161
treatment2:age_c	1	0.9368	0.9368	3.3152	0.0691
Residuals	679	191.8754	0.2826	NA	NA

For a term with 2 level or a continuous predictor, the t-test and F-test both test $H_0 : \beta_j = 0$ For `ethnicity`, there are 3 dummy coefficients. t-test compare each category to the reference group, whereas F-test tests whether at least 1 of the coefficient is different from 0.

3.3 Preference for Chance Choice

Solution

First, we fit the model. The residual analysis in Figure 10 shows that the residuals follow an approximate Gaussian distribution, are centered around 0, and have roughly common variance. Cook's D and Studentized Residual plots in Figure 11 confirm that there are no strongly influential points in our dataset. The multicolinearity check in Table 7 demonstrates that there is no significant multicolinearity. The conditions of use for regression are met.

The model discernibly explains the variation in the outcome, $F_{16,678} = 4.07, p < 0.001, \omega^2 = 0.07$. The effect size is small.

Table 9 shows that the effect of treatment is statistically discernible, $F_{1,678} = 37.19, p < 0.001$, and the interaction effect of Treatment $\times$ Age is also statistically significant, $F_{1,678} = 4.53, p = 0.034$.

We calculate the estimated marginal trends (EMT) to understand which group drives the interaction effect. Table 11 reveals that the slope of age vs. rating is only statistically discernible for the Negative Framing condition, since the confidence interval of the slope of age vs. rating (i.e., how much the rating changes per 1-unit increase in age) in the Positive Framing condition does include 0.

A Johnson-Neyman analysis shows the effect of treatment at each age and highlights at which age(s) this effect is statistically discernible. Particularly, Figure 13 tells us what the slope of the treatment term is at each age (i.e., how much the rating changes per 1-unit increase in treatment at each age). Since the treatment only has 2 levels, a 1-unit increase means a change in framing (positive to negative). The framing difference (negative - positive) is statistically discernible from about age 27 onward.

3.3.1 Fit the model

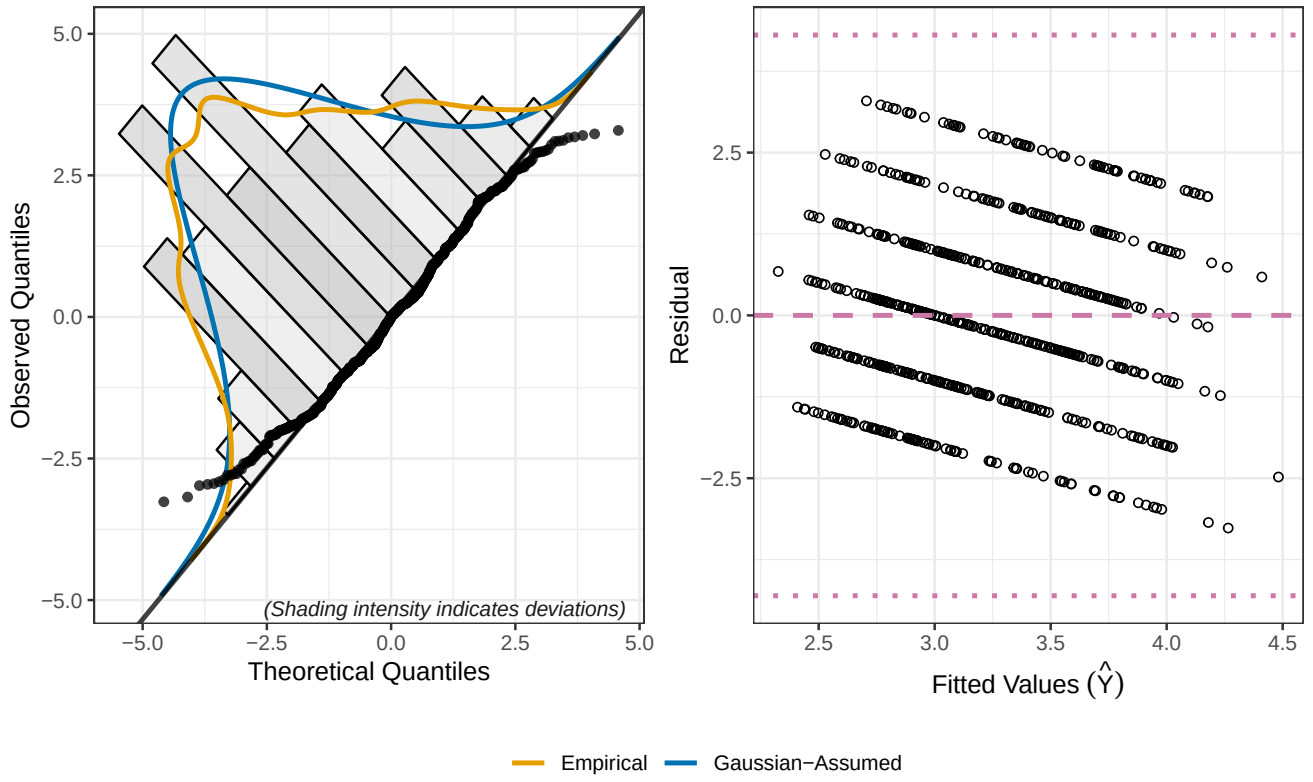
There is 1 `NA` in this column, which `lm()` automatically drops.

```
mod.q3 = lm(pick3 ~ treatment3 * age_c + gender + ethnicity
                  + education + politics,
                  data = dat)
```

3.3.2 Residual and outlier analysis

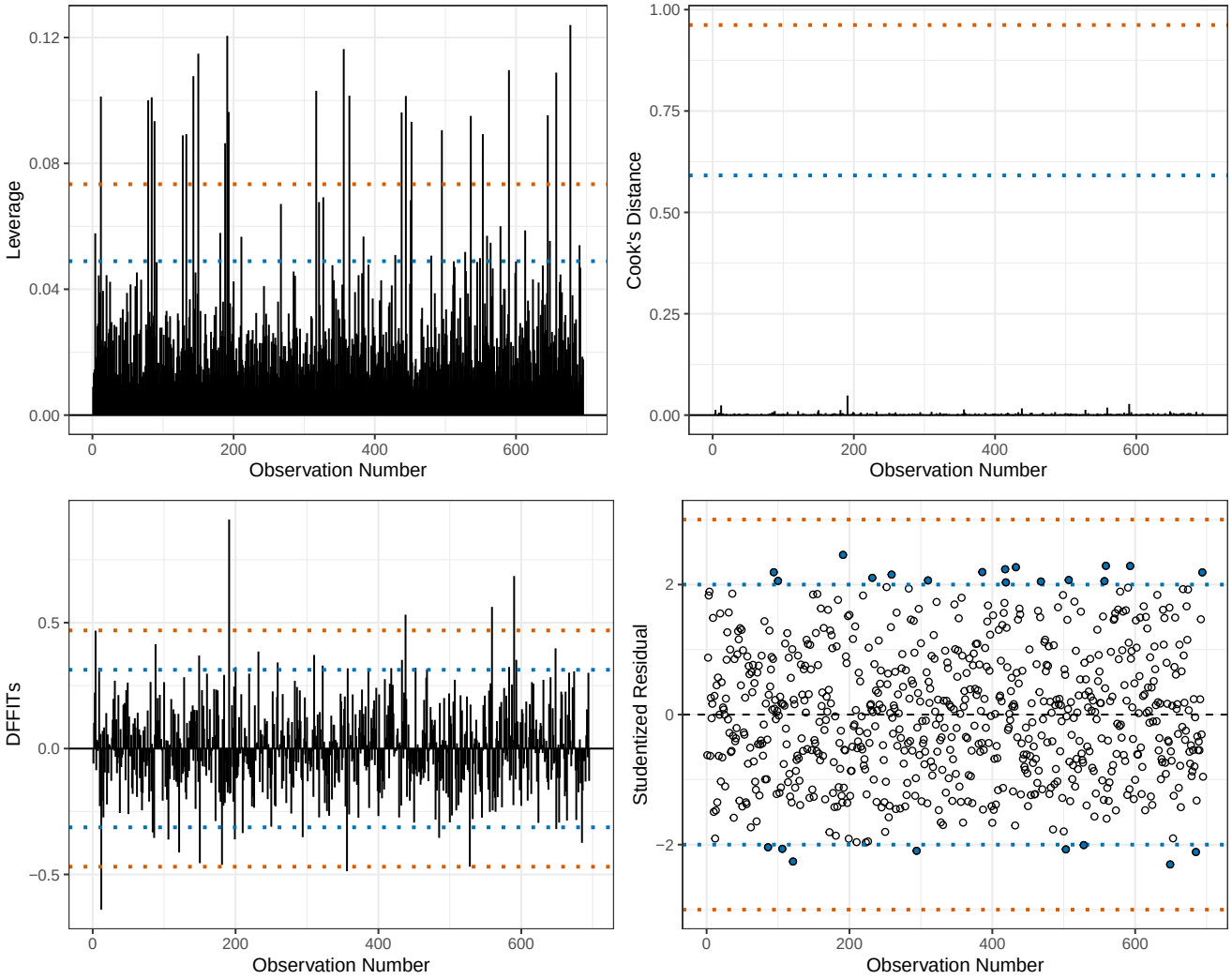
```
plotResiduals(mod.q3)
```

Figure 10: Residual Analysis




```
plotInfluence(mod.q3)
```

Figure 11: Outlider Analysis



3.3.3 Multicolinearity check

Table 7: VIF

	GVIF	Df	GVIF^(1/(2*Df))
treatment3	1.0296	1	1.0147
age_c	2.0203	1	1.4214
gender	1.0826	2	1.0200
ethnicity	1.1321	3	1.0209
education	1.1300	5	1.0123
politics	1.1845	3	1.0286
treatment3:age_c	2.0114	1	1.4182

3.3.4 Overall F-test

```
sum.q3 = summary(mod.q3)
```

```
q3 = as_tibble_row(sum.q3$fstatistic) |>
  rename(F = value, df1 = numdf, df2 = dendf) |>
  mutate(p = pf(F, df1, df2, lower.tail = FALSE),
         omega2 = F_to_omega2(F, df1, df2)$Omega2)
```

```
(interpret_omega_squared(0.07, "cohen1992"))
```

```
[1] "small"
(Rules: cohen1992)
```

Table 8: Overall F test

F	df1	df2	p	omega2
4.0744	16	678	0	0.0661

Table 9: ANOVA

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
treatment3	1	76.5801	76.5801	37.1926	0.0000
age_c	1	2.8446	2.8446	1.3815	0.2403
gender	2	2.3545	1.1773	0.5718	0.5648
ethnicity	3	10.2783	3.4261	1.6640	0.1735
education	5	9.0132	1.8026	0.8755	0.4971
politics	3	23.8242	7.9414	3.8569	0.0094
treatment3:age_c	1	9.3321	9.3321	4.5323	0.0336
Residuals	678	1396.0119	2.0590	NA	NA

3.3.5 Inference about coefficients

```
coefs.q3 = summary(mod.q3)$coefficients
```

Table 10: Inferences about Coefficients

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	3.4482	0.2860	12.0565	0.0000
treatment3pos3	-0.5999	0.1105	-5.4306	0.0000
age_c	0.0168	0.0063	2.6739	0.0077
genderMale	0.1214	0.1133	1.0719	0.2841
genderThird gender/Prefer not to say	-0.0132	0.4140	-0.0318	0.9746
ethnicityBlack or African American	0.0915	0.2769	0.3305	0.7412
ethnicityOther/Prefer not to say	-0.4027	0.3247	-1.2403	0.2153
ethnicityWhite	-0.2164	0.2017	-1.0730	0.2837
education2	0.1152	0.2199	0.5237	0.6007
education3	-0.1765	0.2461	-0.7175	0.4733
education4	0.1355	0.1982	0.6834	0.4946
education5	-0.1289	0.2206	-0.5844	0.5591
education6	0.0010	0.4712	0.0021	0.9983
politicsIndependent	0.4718	0.1352	3.4897	0.0005
politicsOther/Prefer not to say	0.1154	0.2698	0.4277	0.6690
politicsRepublican	0.1821	0.1445	1.2606	0.2079
treatment3pos3:age_c	-0.0191	0.0090	-2.1289	0.0336

3.3.6 Post-hoc testing

```
emt.q3 = emtrends(mod.q3, ~ age_c * treatment3, var = "age_c")
# var: the slope of the continuous variable that you want to compare across groups
```

Table 11: Estimated Marginal Trends

age_c	treatment3	age_c.trend	SE	df	lower.CL	upper.CL
-0.0103	neg3	0.0168	0.0063	678	0.0045	0.0291
-0.0103	pos3	-0.0023	0.0065	678	-0.0151	0.0105

The first column, `age`, is the mean age of the sample (standardized), at which the trends were evaluated.

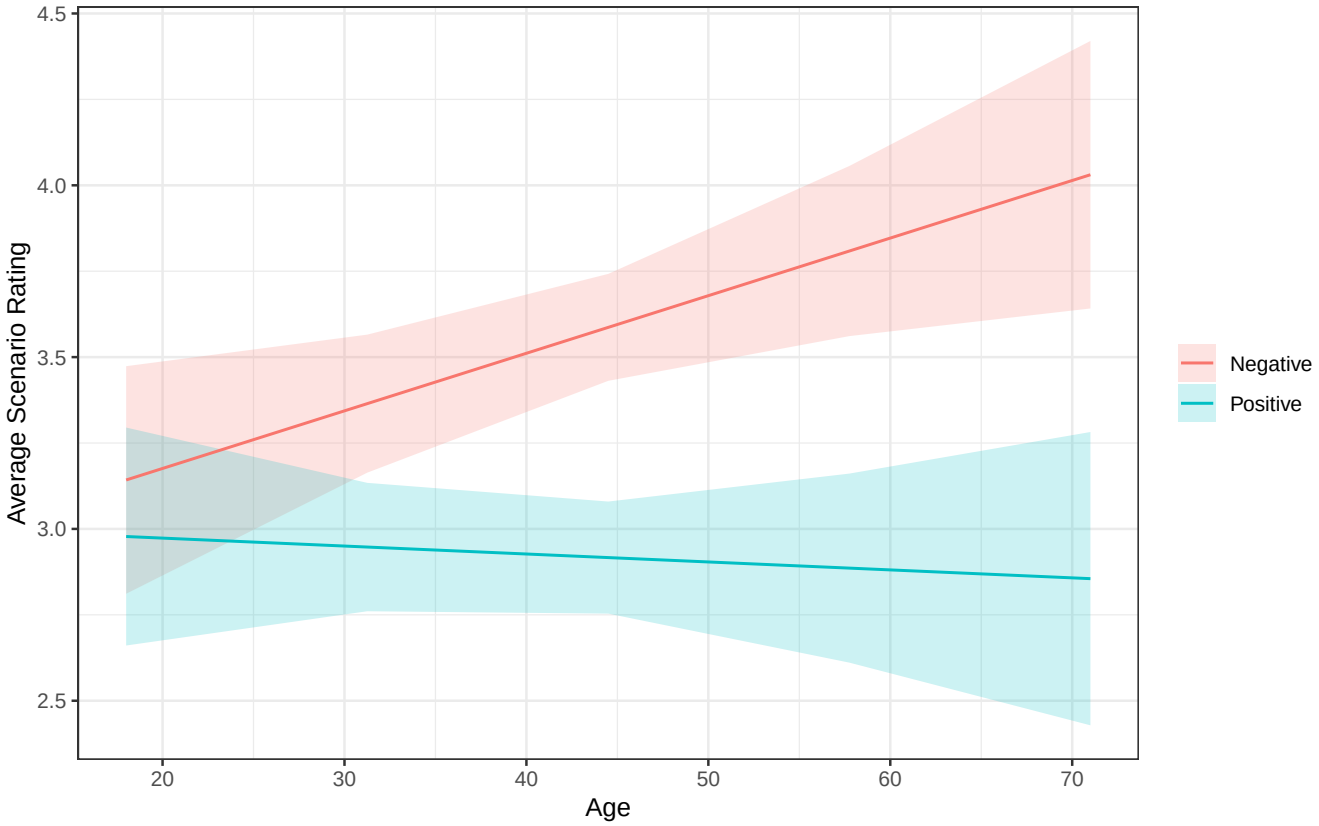
By default, `emmeans` averages over gender, ethnicity, education, and politics by giving every level equal weight, even levels with few people. Setting `weights = "proportional"` makes sure each level is weighted by how common it is in the dataset. For example, if 80% of our sample is White, then the coefficient for that level the categorical predictor is multiplied by an according weight.

```
emm.q3 = emmeans(mod.q3, ~ treatment3 | age_c,
                 at = list(age_c = seq(min(dat$age_c), max(dat$age_c), length.out = 5)),
                 weights = "proportional")

plot.q3 = as.data.frame(emm.q3)  |>
  mutate(treatment3 = case_when(treatment3 == "pos3" ~ "Positive",
                                treatment3 == "neg3" ~ "Negative")) |>
  mutate(age = age_c + mean(dat$age))
# back to real age for x-axis
```

```r
ggplot(plot.q3,
  aes(x = age,
      y = emmean,
      color = treatment3,
      fill = treatment3)) +
  geom_ribbon(aes(ymin = lower.CL,
                  ymax = upper.CL),
                  alpha = 0.2,
                  color = NA) +
  geom_line(linewidth = 0.6) +
  labs(x = "Age",
       y = "Average Scenario Rating", color = NULL, fill = NULL) +
  theme_bw()
```

Figure 12: Participants' Attitudes Towards the Scenario



We transform the variable `treatment 3` to a discrete binary variable (x_1). $x_1 = 0$ when treatment is positive, and $x_1 = 1$ when treatment is negative. The regression model with the interaction term between treatment (x_1) and age (x_2) can be written as:

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2 + C$$

or, singling out x_1,

$$\hat{y} = \beta_0 + (\beta_1 + \beta_3 x_2) x_1 + \beta_2 x_2 + C$$

ANOVA tests whether β_3 is discernibly different from 0. EMT etimates the age slope within each framing group and β_3 is their difference. When `treatment3` is negative, $x_1 = 1$. $\hat{y}$ changes by $\beta_2 + \beta_3$ with every unit change in x_2. When `treatment3` is positive, $x_1 = 0$. $\hat{y}$ changes by β_2 with every unit change in x_2.

The Johnson-Neyman analysis tests the slope of x_1 $(\beta_1 + \beta_3 x_2)$ at every value of x_2. Specifically, it tests, at each age, whether the difference in predicted ratings between negative and positive framing is discernibly different from 0.

```r
dat = dat |> mutate(neg3 = as.numeric(treatment3 == "neg3"))

mod.q3.jn = lm(pick3 ~ neg3 * age + gender + ethnicity + education + politics,
               data = dat)

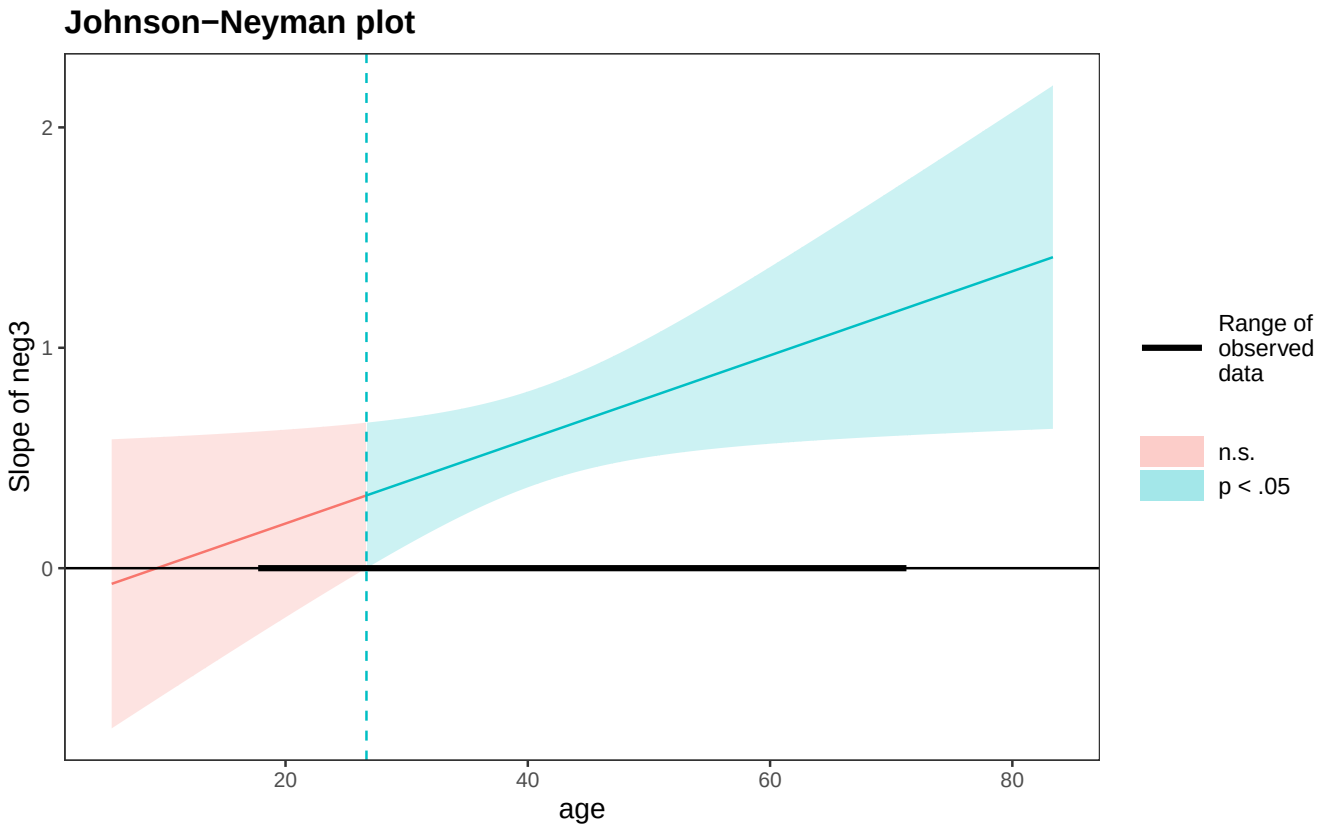
jn = interactions::johnson_neyman(mod.q3.jn, pred = neg3, modx = age, alpha = .05)
```

```r
jn$bounds
```

```
     Lower     Higher
-366.13671   26.68121
```

```r
jn$plot
```

Figure 13: Johnson-Neyman Plot



4 Discussion

There is a statistically discernible effect of framing in all three experiments, confirming previous literature.

Regarding the Age × Framing interaction effect on participants' perception of a scenario, in two out of three experiments, particularly attribute framing and goal framing, we did not find a statistically discernible effect.

In the risky-choice framing, however, the age × framing interaction effect was statistically discernible, $F_{1,678} = 4.53, p = .034$, with a slope difference of 0.019 rating points per year of age (SE = 0.009). This effect has been adjusted for simultaneous linear changes in gender, ethnicity, education, and politics.

Particularly, under negative framing, ratings of the risky option rose by 0.017 points per 1-year increase in age (95% CI [0.005, 0.029]). As adults get older, they are more likely to take risk when the scenario is framed negatively. Ratings under positive framing did not change with age (−0.002 points per year, 95% CI [−0.015, 0.011]). The Johnson-Neyman analysis shows that negative framing induced higher risk-seeking than positive framing for participants older than about 27.

References

Xu, Lance. 2024. "Advanced Age Predicts Increased Susceptibility to Attribute, Goal, and Risky-Choice Framing in Negative Frame Valences." *Humanities and Social Sciences Communications* 11 (1): 217.